5. The age of a dead animal can be determined by using an isotope of _____
   (a) carbon (b) iodine
   (c) phosphorous (d) oxygen
6. Which of the following does not contain natural dyes?
   (a) Potato (b) Beetroot
   (c) Carrot (d) Turmeric
7. This type of food protect us from deficiency diseases.
   (a) Carbohydrates (b) Vitamins
   (c) Proteins (d) Fats
8. Radiochemistry deals with
   (a) oxidants (b) batteries
   (c) isotopes (d) nanoparticles
9. The groups responsible for the colour of an organic compound is called
   (a) isotopes (b) auxochrome
   (c) chromogen (d) chromophore
10. Chlorinated hydrocarbons are used as
    (a) fertilizers (b) pesticides
    (c) food colourants (d) preservatives

II. Fill in the blanks.

1. _____ is an electrochemical cell which converts electrical energy into chemical change(Reaction).
2. Painkiller drugs are called _____
3. Indigo is a _____ dye.
4. _____, _____ and _____ are macronutrients required for plant growth.
5. _____ is a chemical used in finger print analysis.

III. Match the following.

Antipyretics	-	Large surface area
Corrosion prevention	-	Iodine-131
Hyperthyroidism	-	Fever
Nanoparticle	-	Body building
Proteins	-	Electroplating

IV. Answer briefly.

1. What is Radio Carbon Dating?
2. What are called Anaesthetics? How are they classified?
3. What is the need for chemical fertilizers in crop fields?
4. What is Forensic chemistry related to?

V. Answer in detail.

1. Explain the types of dyes based on their method of application.
2. Name various food additives and explain their functions.

VI. Higher Order Thinking Skills.

1. Batteries that are used in mobile phone can be recharged. Likewise, can you recharge the batteries used in watches? Justify your answer.
2. Sudha met with a fire accident. What kind of drug(s), she must take?
3. The soil pH of a crop land is 5. What kind of fertilizers should be used in that land?

REFERENCE BOOKS

1. Nanomaterials and Nanochemistry by Catherine Brechignac
2. Nuclear and Radiochemistry – Fundamentals and applications by Karl Heinrich Lieser
3. Food Chemistry (Third Edition) by Owen Fennema

INTERNET RESOURCES

https://en.wikipedia.org/wiki/Agricultural_chemistry

https://www.medicalnewstoday.com/articles/321108.php

https://www.youtube.com/watch?v=kC1aPOqoYWc

IX_SCI_EM_Unit-16_CHE.indd 197 30/11/2022 20:00:46